Applied Numerical Methods for Civil Engineering

CGN 3405 - 0002

Week 7: Nonlinear Equations

Xinyu Chen

Assistant Professor

University of Central Florida

Quizzes Now!

- **Today's participation** (ungraded survey): Please check out

  "`Class Participation Quiz 13`"

  Time slot: **2:30PM – 3:00PM**

  on Canvas.

- **Grading policy**: Given the scores of Exam 1 and Exam 2 as a and b, respectively, only in the case of $b > a$, then your score for both exams will become b.

Exam 1: 75; Exam 2: 90 $\Rightarrow$ Exam 1: 90; Exam 2: 90

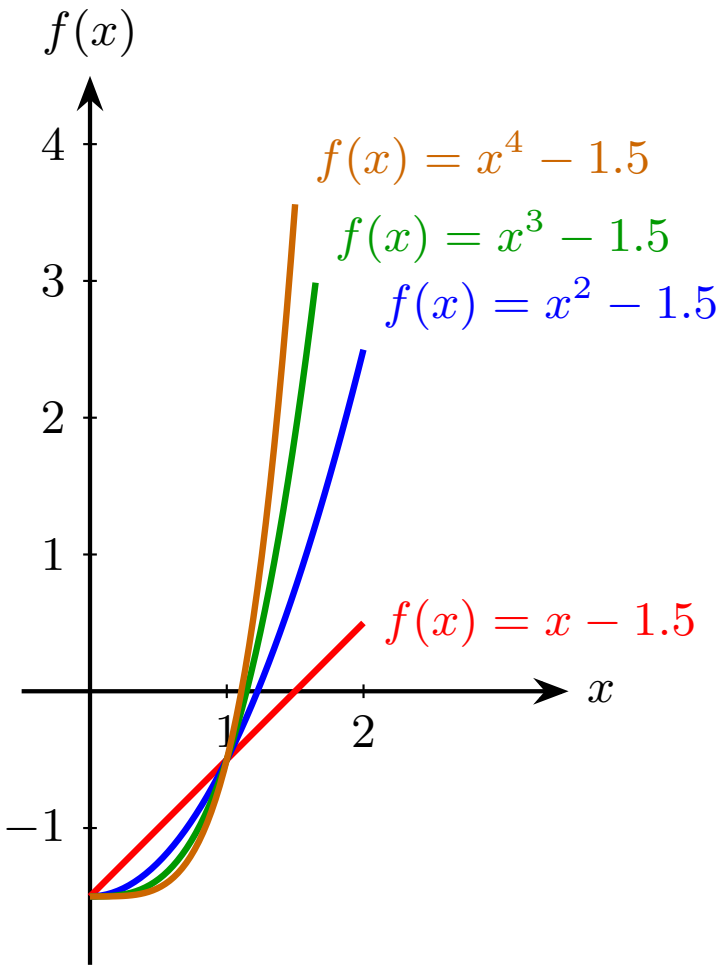
Learning Objectives

By the end of week, you will be able to:

- Derive the Newton-Raphson iteration formula from a Taylor series
- Apply the method to find roots of nonlinear equations
- Analyze the convergence behavior of the method
- Identify situations where the method may fail
- Implement the algorithm with Python programming

Problem Statement

- A nonlinear equation $f(x) = 0$
- Examples:
  - $x^2 - 2 = 0$ (Find $\pm\sqrt{2}$)
  - $\sin(x) = 0$ (Find multiples of π)
  - $e^{-x} - x = 0$ (Intersection of functions)
- Many engineering problems reduce to finding roots!



Historical Development

The modern method is a synthesis of three major mathematical contributions:

- **Isaac Newton (1669)**: Described method for polynomial equations
- **Joseph Raphson (1690)**: Formalized the iterative process
- **Thomas Simpson (1740)**: Generalized to calculus form using derivatives

Derivation from Taylor Series

- Expand $f(x)$ around x_n using Taylor series:

$$
\begin{aligned}
f(x_{n+1}) \approx & f(x_n) \\
& + f'(x_n)(x_{n+1} - x_n) \\
& + \frac{f''(x_n)}{2!}(x_{n+1} - x_n)^2 \\
& + \cdots \\
& + \frac{f^{(t)}(x_n)}{t!}(x_{n+1} - x_n)^t \\
= & \sum_{k=0}^{t} \frac{f^{(k)}(x_n)}{k!}(x_{n+1} - x_n)^k
\end{aligned}
$$

Derivation from Taylor Series

- Expand $f(x)$ around x_n using Taylor series:

$$
\begin{aligned}
f(x_{n+1}) \approx & f(x_n) \\
& + f'(x_n)(x_{n+1} - x_n) \\
& + \frac{f''(x_n)}{2!}(x_{n+1} - x_n)^2 \\
& + \cdots \\
& + \frac{f^{(t)}(x_n)}{t!}(x_{n+1} - x_n)^t \\
= & \sum_{k=0}^{t} \frac{f^{(k)}(x_n)}{k!}(x_{n+1} - x_n)^k
\end{aligned}
$$

- We want $f(x_{n+1}) = 0$
- Only use the first-order approximation:

$$f(x_{n+1}) = 0 \quad \Rightarrow \quad f(x_n) + f'(x_n)(x_{n+1} - x_n) \approx 0$$

Newton-Raphson Iteration Formula

- Let the first-order approximation be 0:

$$f(x_n) + f'(x_n)(x_{n+1} - x_n) = 0$$

- Rearrange the equation to solve for x_{n+1}:

$$\begin{aligned} & f'(x_n)(x_{n+1} - x_n) = -f(x_n) \\ \Rightarrow \quad & x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \\ \Rightarrow \quad & x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \end{aligned}$$

which is the Newton-Raphson iteration formula.

Newton-Raphson Iteration Formula

Understanding the formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- x_n: Current estimate
- $f(x_n)$: Function value at current estimate
- $f'(x_n)$: Slope (first-order derivative) at current estimate
- $\frac{f(x_n)}{f'(x_n)}$: Correction term

Intuition: If $f(x_n)$ is large, we take a big step. If slope is steep, we take a smaller step.

Newton-Raphson Method

- Choose initial point x_0 and tolerance ε
- Set $n = 0$
- Repeat until convergence (using `while`)
  - Compute $f(x_n)$ and $f'(x_n)$
  - Calculate new estimate:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

  - Check convergence: If $\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \varepsilon$, stop
  - Set $n = n + 1$

Stopping Criteria

Three common stopping criteria:

- Absolute error

$$|x_{n+1} - x_n| < \varepsilon$$

- **Relative error**

$$\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \varepsilon$$

- Residual

$$|f(x_{n+1})| < \varepsilon$$

assume $f(x) = 0$ for some x

In practice, use a combination for robustness

Stopping Criteria

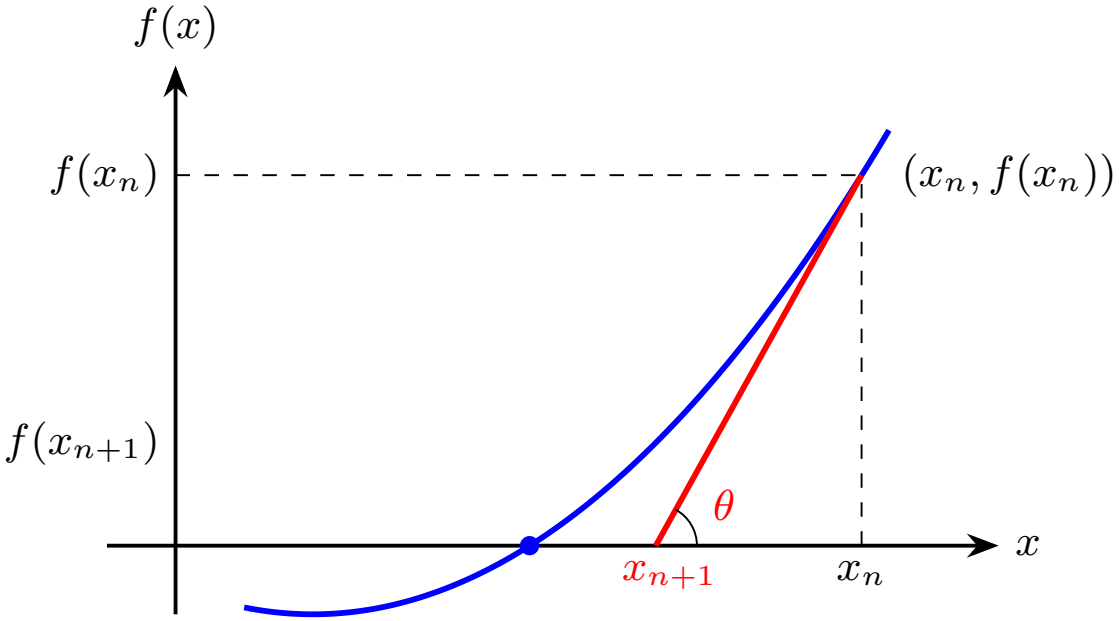
Always include a **maximum iteration limit** to prevent infinite loops

```
iter = 0
max_iter = 100
while error > tol and iter < max_iter:
    # Newton-Raphson iteration formula
    iter += 1
if iter >= max_iter:
    print('Warning: Maximum iterations reached')
```

Intuitive Understanding of Newton-Raphson Method

Given the **starting point** x_n for finding the root of the equation $f(x) = 0$

- Locate the point on the curve at this guess: $(x_n, f(x_n))$
- Draw the **tangent line** to the curve at this point
- Observe where this **tangent line crosses the x-axis**: $(x_{n+1}, 0)$

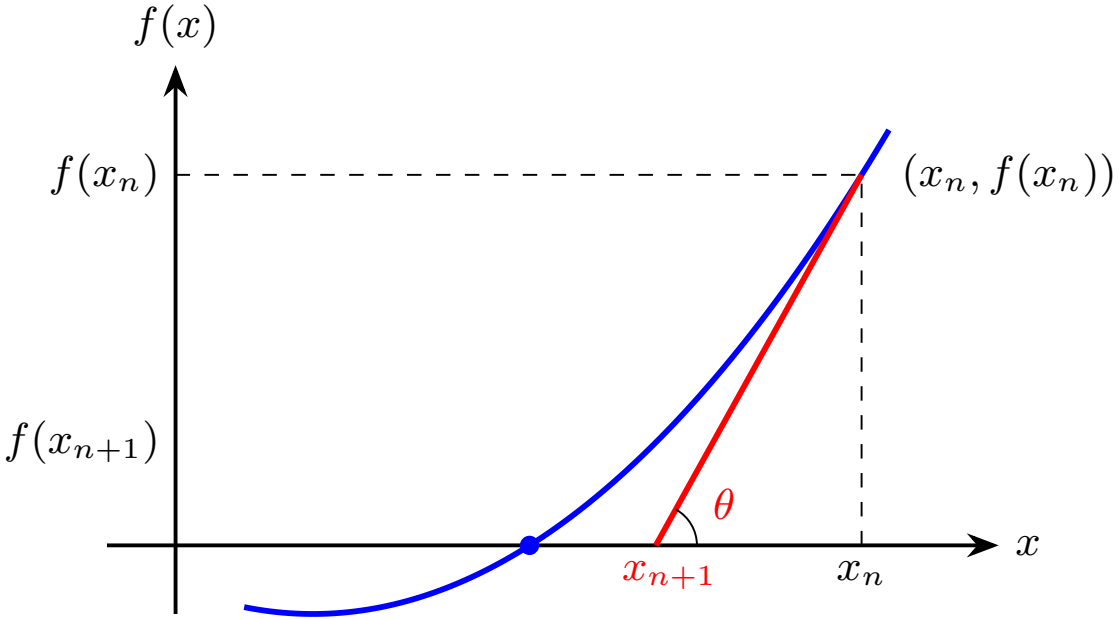


Intuitive Understanding of Newton-Raphson Method

Slope $f'(x_n)$:

- Derivative of function $f(x)$ evaluated at x_n
- Definition of the slope of a line:

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{0 - f(x_n)}{x_{n+1} - x_n}$$



Intuitive Understanding of Newton-Raphson Method

We start with the equation:

$$f'(x_n) = \frac{-f(x_n)}{x_{n+1} - x_n}$$

1. Multiply both sides by $(x_{n+1} - x_n)$:

$$f'(x_n) \cdot (x_{n+1} - x_n) = -f(x_n)$$

2. Divide both sides by $f'(x_n)$:

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)}$$

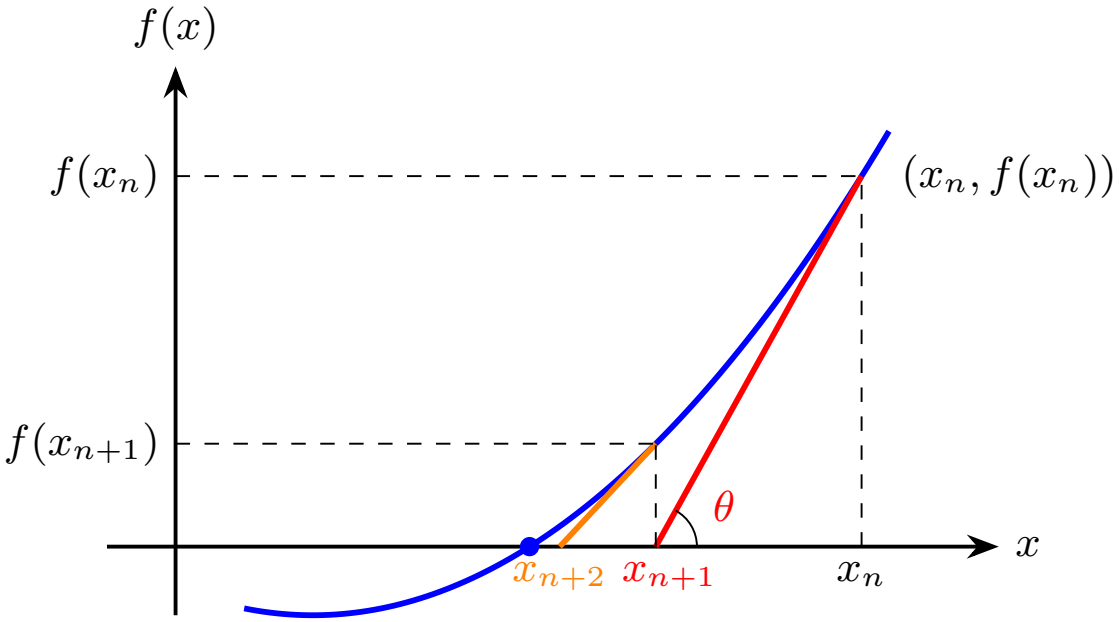
3. Add x_n to both sides to isolate x_{n+1}:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Intuitive Understanding of Newton-Raphson Method

How about slope $f'(x_{n+1})$?

$$f'(x_{n+1}) = \frac{0 - f(x_{n+1})}{x_{n+2} - x_{n+1}}$$



Intuitive Understanding of Newton-Raphson Method

How about slope $f'(x_{n+1})$?

$$f'(x_{n+1}) = \frac{0 - f(x_{n+1})}{x_{n+2} - x_{n+1}}$$

1. Multiply both sides by $(x_{n+2} - x_{n+1})$:

$$f'(x_{n+1}) \cdot (x_{n+2} - x_{n+1}) = -f(x_{n+1})$$

2. Divide both sides by $f'(x_{n+1})$:

$$x_{n+2} - x_{n+1} = -\frac{f(x_{n+1})}{f'(x_{n+1})}$$

3. Add x_{n+1} to both sides to isolate x_{n+2}:

$$x_{n+2} = x_{n+1} - \frac{f(x_{n+1})}{f'(x_{n+1})}$$

Solving $x^2 - 2 = 0$

Given initial point $x_0 = 1.5$, solve $x^2 - 2 = 0$

- Function $f(x) = x^2 - 2$
- Derivative $f'(x) = 2x$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}$$

Solving $x^2 - 2 = 0$

Given initial point $x_0 = 1.5$, solve $x^2 - 2 = 0$

- Function $f(x) = x^2 - 2$
- Derivative $f'(x) = 2x$
- Newton-Raphson iteration formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n} = \frac{x_n}{2} + \frac{1}{x_n}$$

- Newton-Raphson method:
  - First iteration: $x_1 = \frac{x_0}{2} + \frac{1}{x_0} = 1.416666667$
  - Second iteration: $x_2 = \frac{x_1}{2} + \frac{1}{x_1} = 1.414215686$
  - Third iteration: $x_3 = \frac{x_2}{2} + \frac{1}{x_2} = 1.414213562$
  - Fourth iteration: $x_4 = \frac{x_3}{2} + \frac{1}{x_3} = 1.414213562$

Converges to $\sqrt{2} = 1.414213562$ in just **3 iterations**!

Solving $x^2 - 2 = 0$

```python
def newton_raphson_sqrt2(x0, tol=1e-9, max_iter=100):
    n = 0
    xn = x0
    while n < max_iter:
        # Calculate f(xn) and f'(xn)
        fx = xn**2 - 2
        f_prime = 2 * xn
        # Newton-Raphson Formula
        x_next = xn - (fx / f_prime)
        # Calculate Relative Error
        error = abs(x_next - xn) / abs(x_next)
        # Check Convergence
        if error < tol:
            return x_next
        xn = x_next
        n += 1
    return xn

# Run with initial point x0 = 1.5
root = newton_raphson_sqrt2(1.5)
```

Quick Summary

Monday's Class:

- Newton-Raphson iteration formula: Derivation from Taylor series
- Intuitive understanding of Newton-Raphson method
- Solve the nonlinear equation $x^2 - 2 = 0$